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Café Customer Analytics & Churn Prediction

Turning transaction data into customer insight, churn prediction, and retention action.

Data Analytics / Machine Learning Portfolio · September 2026

Portfolio case study based on course-provided café transaction data.

Full reproducible analysis is included in the accompanying Colab notebook.

What I built and why it matters

A concise case study, not just a collection of code outputs

Project goal

Analyze café transactions to identify why revenue changed, which customers are drifting away, and how retention actions could be prioritized.

My contribution

- Rebuilt the workflow from the course datasets
- Cleaned and engineered features for analysis
- Designed leakage-aware churn labels
- Compared models with cross-validation
- Converted predictions into action priorities

Portfolio framing

The Colab notebook is the reproducible code. This deck is the curated case study for application review.

Scope note

The dataset is educational. Recommendations are hypotheses and should be validated with A/B testing before real deployment.

Deliverables

PPT/PDF

portfolio summary

Colab

full analysis

GitHub

review package

Revenue decline was an activity problem

Average order value was stable, while orders and active customers fell



-20.6%

Jan → Dec revenue

-19.4%

order volume

-12.6%

active customers

-1.4%

average order value

Interpretation

The main problem is not that customers spend much less per order. The bigger issue is that fewer customers and orders remain active.

Decision direction

Retention and reactivation are more relevant than blanket discounting on every transaction.

I investigated the problem through four connected lenses

The workflow moves from business diagnosis to operational action

1

Business trend

Revenue, orders, active customers, AOV

3

Segmentation

RFM-style K-Means with PCA visualization

2

Customer value

Pareto curve and visit-frequency behavior

4

Prediction

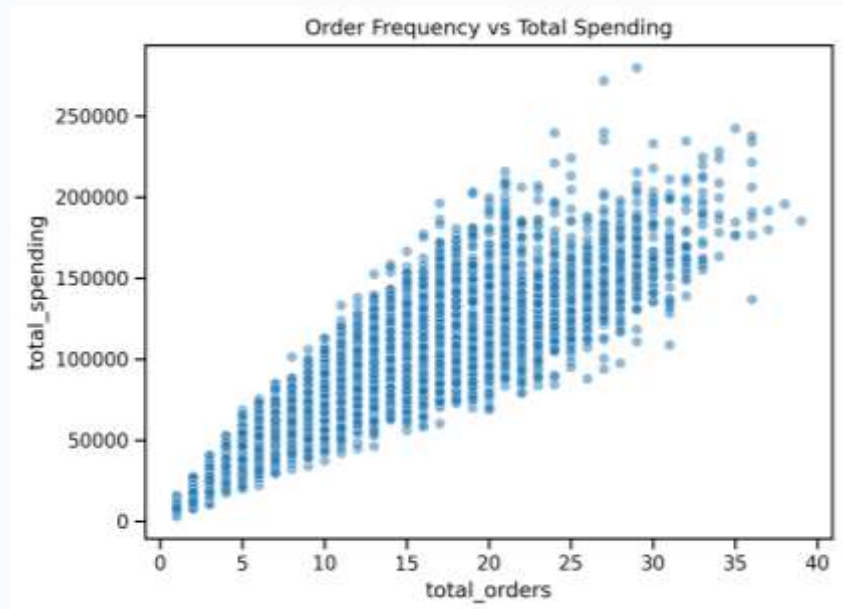
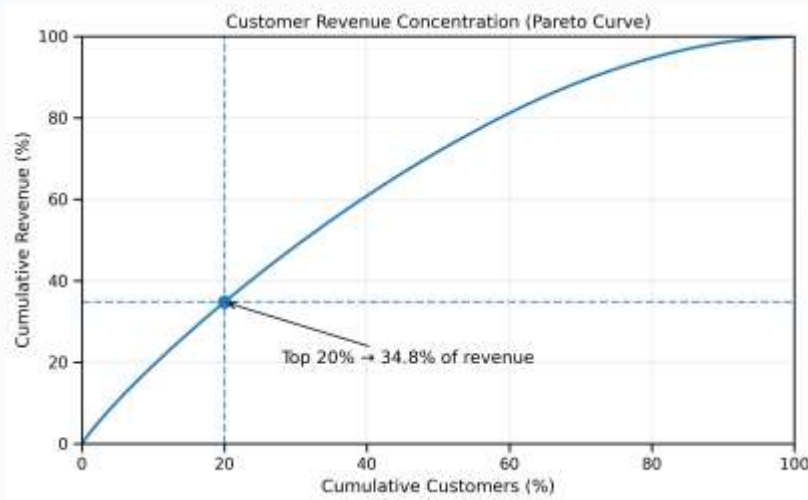
Leakage-aware churn model with cross-validation

Why this structure works

The project does not stop at model accuracy. It connects the business issue to customer groups, churn risk, and retention prioritization.

Customer value is not extreme 80/20, but visits clearly matter

The top 20% contributed 34.8% of revenue, and spending rises with frequency



34.8%

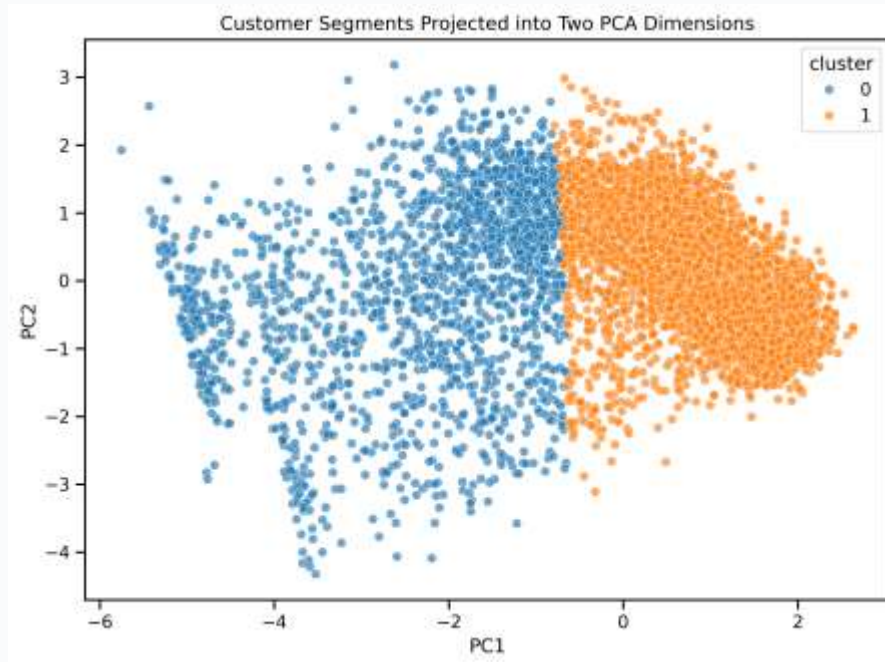
revenue from top customer quintile

Key decision

Focus on increasing repeat visits across segments instead of assuming value comes only from a tiny group of super-customers.

K-Means separated active customers from drifting customers

RFM-style behavior features were standardized and visualized through PCA



Inactive / At-Risk

1,721

customers

175

avg recency days

4.1

avg orders

Active / Loyal

4,279

customers

25

avg recency days

17.4

avg orders

Portfolio insight

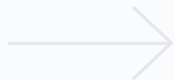
The segments translate raw transaction history into a practical retention lens: who is engaged, and who is drifting away.

The target was designed to avoid future-data leakage

The model predicts Oct–Dec return behavior using only Jan–Sep features

Observation period

January–September 2025
Features known by Sept 30



Outcome period

October–December 2025
Did the customer return?

Inputs used

- Recency, frequency, monetary value
- Average order value and active months
- Channel behavior and coupon share
- Age, membership, store type, source, category

19.3%

observed churn rate

5-fold

stratified cross-validation

20%

held-out test set

My modeling choice

I separated the feature window and outcome window so the model would be evaluated as if it were making a real future prediction.

I selected Logistic Regression for performance and interpretability

Random Forest was similar, but Logistic Regression was stronger in cross-validation

Selected model

0.965

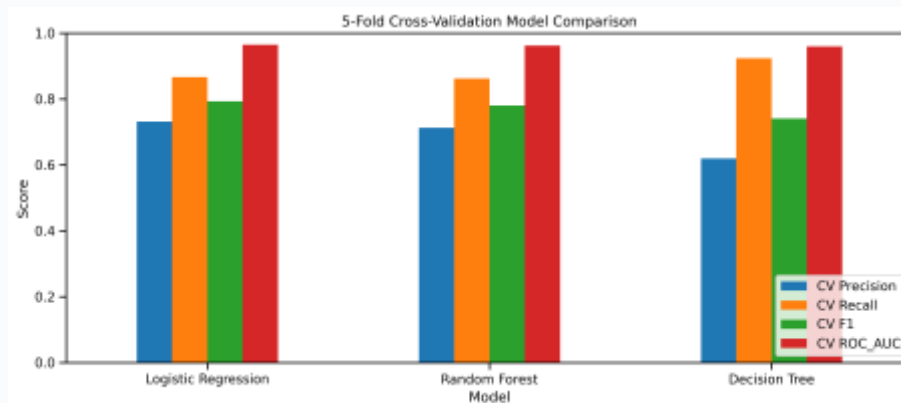
CV ROC-AUC

0.959

test ROC-AUC

0.819

test recall



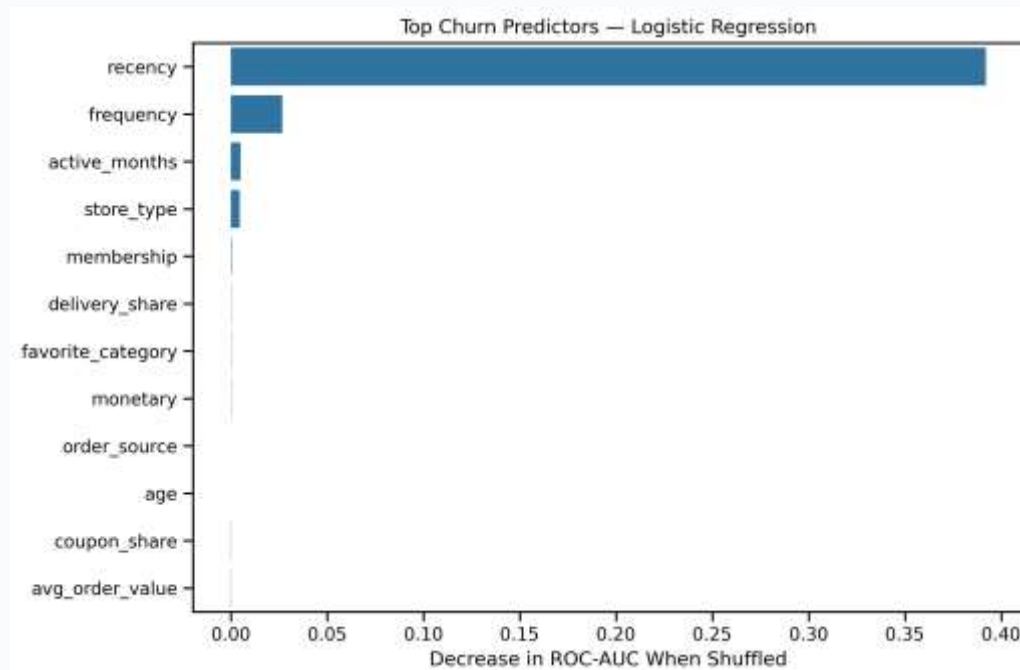
Why this is stronger than just using the complex model

The selected model is accurate enough and easier to explain. That matters because retention decisions affect real customers and budgets.

Decision criterion: cross-validated ROC-AUC, with F1 and recall used as supporting metrics.

Last purchase timing was the strongest churn signal

Permutation importance shows that recency dominates the prediction



0.392

recency importance

Main interpretation

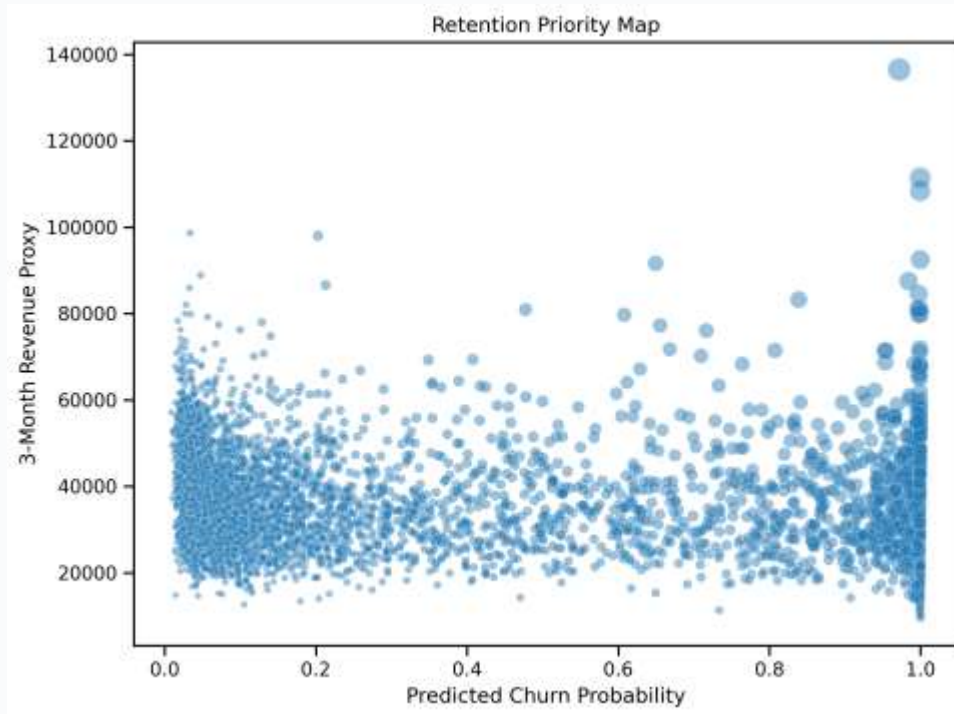
A customer who has not purchased recently is much more likely to be at risk than a customer with similar profile fields but recent activity.

Business takeaway

Retention should trigger early when recency starts rising, not only after a customer has fully disappeared.

Predictions were converted into a prioritized customer list

Risk alone is not enough; I combined churn probability with value-at-risk



600

priority customers

How the list was built

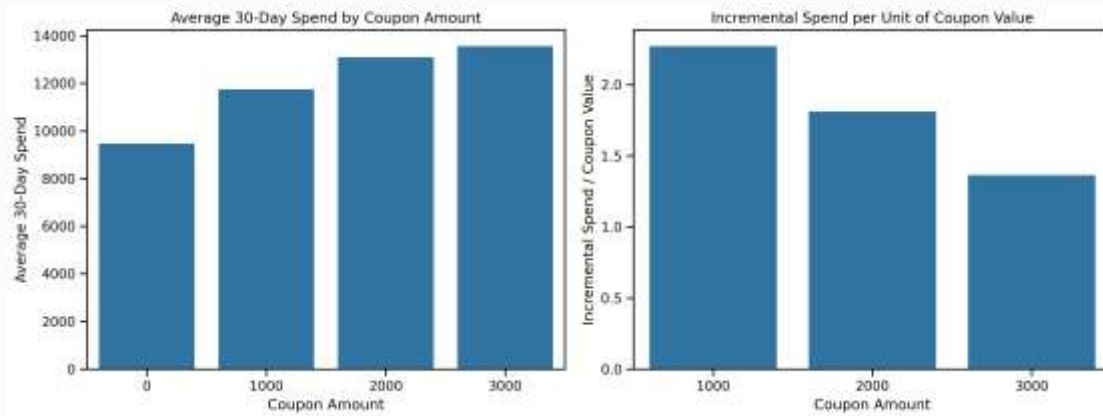
- Estimate churn probability
- Estimate three-month value proxy
- Multiply risk by value
- Rank customers for retention outreach

Why this matters

A practical model should tell the business where to start. This ranking turns probability into a campaign planning tool.

Bigger discounts raised spend, but smaller coupons were more efficient

The coupon pilot separates absolute uplift from efficiency per coupon value



+43.2%

observed uplift with 3,000 coupon

2.27×

incremental spend per 1,000 coupon value

1,500

customers per pilot group

Causal note: treated as descriptive unless random assignment is confirmed.

Potential strategy

Test higher coupons for high value-at-risk customers; use smaller offers when efficiency is the priority.

This project shows an end-to-end analytics workflow

From raw transactions to segmentation, prediction, and retention recommendations

What this portfolio demonstrates

- I diagnosed the business issue using trend decomposition.
- I segmented customers using behavior instead of averages.
- I built a leakage-aware churn model and evaluated it properly.
- I translated predictions into a retention priority list.
- I considered incentive efficiency, not just spend uplift.

Limitations

- Educational dataset
- Coupon result is descriptive unless randomization is confirmed
- Real deployment would require margin and A/B testing

Review package

Colab Notebook

Full reproducible Python analysis with exported outputs.

GitHub Repository

Notebook, README, data notes, and output summaries for independent review.

Portfolio PDF

This deck exported as PDF for application review.

Add final GitHub / Colab links here before submission.

Thank you for reviewing my project.